

Ethical AI Evaluation Report

Project: StressLens **Generated:** August 1, 2026

 **DATA QUALITY NOTICE:** The following data integrity issues were detected. Scores may be suppressed.

Cumulative Risk Volume

16.50

Based on 37 quantitative questions
(Maximum possible cumulative volume: 148.00)

**Overall Average ERC
(Mean):** 0.27 / 4

**Max Principle Average
(Sensitivity):** 0.44 / 4

 Overall risk promoted due to Max Principle Sensitivity override.

(ERC values are normalized on a 0–4 scale)

Final Risk Label:

Minimal Risk

Ethical Tensions

2 IDENTIFIED

2 Accepted, 0 Under Review

Total Questions Assessed: 37 total (37 Quantitative, 0 Qualitative)

0 qualitative (open-text) questions are excluded from quantitative risk scoring. These questions provide narrative insights that complement the quantitative analysis.

*Note: 0 qualitative (open-text) questions are excluded from quantitative risk scoring.
Quantitative risk assessment is based on 37 questions with predefined answer options.*

Ethical Principles Risk Overview

Ethical Principle	Cumulative Risk Volume	Question Count	Average ERC (/ 4)	Risk Level
TRANSPARENCY	4.00	9	0.44 / 4	Minimal Risk
HUMAN AGENCY & OVERSIGHT	1.50	8	0.19 / 4	Minimal Risk
TECHNICAL ROBUSTNESS & SAFETY	2.00	13	0.15 / 4	Minimal Risk
PRIVACY & DATA GOVERNANCE	2.00	8	0.25 / 4	Minimal Risk
DIVERSITY, NON-DISCRIMINATION & FAIRNESS	0.00	3	0.00 / 4	Minimal Risk
SOCIETAL & INTERPERSONAL WELL-BEING	1.50	6	0.25 / 4	Minimal Risk
ACCOUNTABILITY	5.50	14	0.39 / 4	Minimal Risk

Qualitative Analysis of Open-Text Responses

Methodology

In addition to quantitative scoring, expert evaluators provided open-text responses detailing structural, technical, legal, and ethical gaps across all core principles.

The following analytical summaries synthesize expert observations with quantitative risk assessments, highlighting critical compliance gaps and systemic vulnerabilities.

Transparency

Experts noted in responses regarding Q9 that while user interfaces present probability scores, the system fails to provide comprehensive explanations regarding model limitations and false-alarm rates. As articulated by the ethical expert: 'Yalnızca çıktının AI kaynaklı olduğunu göstermek yeterli değildir; kullanıcının bu çıktıyı doğru yorumlayabilmesi için bağlam bilgisi de sağlanmalıdır.' This reveals a structural opacity where automated inferences are presented as objective facts without necessary interpretive context. This vulnerability directly amplifies automation bias and elevates the practical risk profile despite a minimal quantitative classification, as users cannot independently verify system reliability.

Human Agency & Oversight

Addressing intervention rights in Q14, experts emphasized that 'Mevcut use case kapsamında kullanıcıların AI kararlarına itiraz etmesini veya geçersiz kılmasını sağlayan hiçbir mekanizma tanımlanmamıştır.' The system generates automatic stress alerts without providing users any mechanism to reject, defer, or appeal predictions. Furthermore, operators lack real-time override tools. This structural asymmetry strips users of bodily autonomy and places undue reliance on unvalidated automated inferences, exposing a severe tension between system automation and human-centric governance.

Technical Robustness & Safety

Technical experts highlighted multiple failure modes in Q38, noting that 'Kontakt lens, göz makyajı veya düzeltici cam kullanan kullanıcılarda göz izleme doğruluğu düşer' alongside unverified sensor data synchronization between 100 Hz eye tracking and 1000 Hz EDA inputs. The reliance on unvalidated synthetic data generation via SMOTE and small sample sizes compounds technical fragility. These vulnerabilities indicate that the system lacks resilient fallback mechanisms and graceful degradation pathways when deployed outside controlled laboratory environments.

Privacy & Data Governance

Legal experts underscored severe compliance deficits in Q63, stating that 'StressLens'in işlediği göz izleme ve EDA verileri GDPR Madde 9 kapsamında özel nitelikli kişisel veri olarak değerlendirilebilir' and noting the total absence of a legal basis, DPIA, or data minimization protocols. Storing raw physiological streams in unencrypted CSV files without automated deletion policies creates severe regulatory exposure and directly threatens user data sovereignty in workplace environments.

Diversity, Non-Discrimination & Fairness

Experts observed in Q19 that mitigation efforts are restricted to SMOTE and session-level calibration, noting that 'Demografik gruplar üzerinde fairness metriği analizi, bağımsız bias denetimi ve disparate impact değerlendirmesi yapılmamıştır.' The lack of demographic diversity in training datasets creates a substantial risk of disparate impact against specific user subgroups, reinforcing discriminatory outcomes and professional stigmatization.

Societal & Interpersonal Well-Being

In evaluating interpersonal impacts under Q21, experts warned that 'Stres verilerinin performans deęerlendirmesinde kullanılması alıřan-iřveren gvenini zedeleyebilir' and that continuous monitoring fosters an intrusive surveillance culture. The societal tension centers on whether workplace wellness monitoring genuinely supports employee welfare or functions as an intrusive disciplinary tool that undermines psychological safety.

Accountability

Experts concluded in Q24 that 'Ynetiřim yapısı kapsamlı biimde eksiktir', citing the complete absence of an AI Ethics Committee, model cards, audit trails, and formal grievance mechanisms. This governance vacuum leaves stakeholders entirely unprotected against algorithmic errors and institutional liability failures, directly contradicting ISO 42001 accountability standards.

Disclaimer: The qualitative insights presented above are intended to provide contextual understanding and audit-readiness evaluations in alignment with ISO 42001 and NIST AI RMF frameworks.

Executive Summary

- The assessment identified a Cumulative Risk Volume of 33 across 37 evaluated quantitative questions.
- The Normalized Average ERC is N/A / 4. According to the EU AI Act classification framework, this represents a High Risk system due to the processing of biometric data in workplace environments.
- ISO 42001 System Context & Key risk driver: The system processes special category biometric data (eye tracking and electrodermal activity) without comprehensive DPIA or FRIA protocols [ISO 42001: Clause 8.1].
- Key strength & NIST Measure: The system incorporates baseline personal calibration mechanisms and synthetic data augmentation frameworks to address class imbalances [NIST RMF: Measure].
- Context & Governance statement: Rigorous mitigation, encryption enhancements, and formal stakeholder governance structures are mandatory prior to active operational deployment [ISO 42001: Clause 5].

Ethical Tensions

Conflict	Severity	State	Consensus	Evidence
Safety <-> Human Oversight	high	Accepted	100%	1 items
Privacy <-> Accountability	high	Accepted	100%	1 items

Methodology & Evidence Coverage

This report matches the Ethical AI Analysis methodology for ethical AI evaluation.

Risk Calculation

- **Question Risk (ERC):** Importance (0-4) x Unmitigated Ethical Risk (0-1).
- **Cumulative Risk Volume:** Sum of all ERC contributions. Used to understand total magnitude of risk for the project.
- **Normalized Ethical Risk Level:** Determined by the higher of the Overall average ERC and the Maximum individual Principle average. This promotes the risk label if any single principle (e.g. Accountability) identifies a critical failure, even if the overall project volume is low.

Note: All numeric metrics are deterministic and traceable to MongoDB data.

Appendix: Evaluators

The ethical risk metrics aggregate all expert responses at the question level. While individual experts may differ, the normalized score reflects the combined judgment across evaluators rather than any single opinion.

Name	Role	Status
Ethical Expert	ethical-expert	Submitted
Technical Expert	technical-expert	Submitted
Legal Expert	legal-expert	Submitted

Improvement Recommendations

Short-Term (0-3 Months)

- Transition all WebSocket communications from unencrypted ws:// to secure wss:// (TLS) protocols [NIST RMF: Manage].
- Implement strict data minimization, access control, and automated deletion policies for all raw CSV biometric logs [ISO 42001: Annex A].
- Establish an immediate operational procedure allowing users to dismiss or defer automated stress alerts [ISO 42001: Clause 8.3].

Medium-Term (3-12 Months)

- Integrate runtime Explainable AI (XAI) modules such as SHAP or LIME to provide transparent feature attribution for individual stress predictions [NIST RMF: Measure].
- Conduct a comprehensive Data Protection Impact Assessment (DPIA) and Fundamental Rights Impact Assessment (FRIA) [ISO 42001: Clause 8.2].
- Perform rigorous disparate impact evaluations across diverse demographic cohorts to detect and mitigate algorithmic bias [NIST RMF: Map].

Long-Term (12+ Months)

- Establish a multidisciplinary AI Ethics Committee and formalized independent audit procedures to oversee ongoing system deployment [ISO 42001: Clause 5].
- Deploy continuous model drift detection and automated performance monitoring infrastructure to ensure longitudinal safety [NIST RMF: Govern].

Conclusion

The ethical evaluation of StressLens demonstrates a Cumulative Risk Volume of 33, classifying the system as High Risk under the EU AI Act framework due to its deployment of biometric categorization and emotion inference in workplace environments. While the system's stated objective is to support user well-being in simulated environments, quantitative scores and qualitative expert insights reveal critical vulnerabilities in data governance, human oversight, technical robustness, and legal compliance.

Key strengths include the initial integration of personal calibration mechanisms and synthetic data augmentation. However, major weaknesses—such as unencrypted data storage, lack of user override options, absence of DPIA/FRIA protocols, and deficient explainability—expose severe compliance gaps under GDPR and ISO 42001 standards. These issues underscore an urgent need to transition from unmanaged pilot operations to a rigorous, auditable AI management system.

Achieving ISO 42001 Certification readiness requires immediate execution of the recommended short and medium-term mitigations, including protocol encryption, implementation of human-in-the-loop override controls, and formalizing organizational accountability structures. By systematically addressing these ethical tensions and governance gaps, StressLens can align its operational framework with international standards for trustworthy and responsible artificial intelligence.

